Question

The FeSe-based system is the simplest type of iron-based superconducting material. The filling of iron vacancies with Cu in the form of a univalent cation will affect the structure and electrical and magnetic properties of the superconductor. In this experiment, precipitation gravimetry and EDTA complexometric titration were used to determine the chemical composition of a $\mathrm{Fe-Cu-Se}$ superconductor. The specific experimental procedure is as follows:

(a) Accurately weigh approximately $0.4639\,\mathrm{g}$ of the sample into a $100\,\mathrm{mL}$ beaker, add $6\,\mathrm{mL}$ of concentrated nitric acid and $6\,\mathrm{mL}$ of concentrated hydrochloric acid, and heat. Add concentrated hydrochloric acid during heating until the sample is completely dissolved. Add $20\,\%$ hydroxylamine hydrochloride solution dropwise under stirring and heating until precipitation is complete, keep warm for a period of time, carefully filter out the precipitate with pre-weighed filter paper, wash several times with deionized water, dry at low temperature until constant weight, and record the mass as $0.2565\,\mathrm{g}$.

(b) Weigh approximately $0.8527\,\mathrm{g}$ of the sample into a $100\,\mathrm{mL}$ beaker, add $10\,\mathrm{mL}$ of concentrated nitric acid and $10\,\mathrm{mL}$ of concentrated hydrochloric acid, and heat. Add concentrated hydrochloric acid during heating until the sample is completely dissolved. Concentrate the solution to about $10\,\mathrm{mL}$, and dilute to $100\,\mathrm{mL}$ with deionized water. Transfer $25.00\,\mathrm{mL}$ of the above solution to a conical flask, and adjust the pH to $3\sim4$ with $1:1$ ammonia water; add $5\,\mathrm{g}$ of solid ammonium fluoride, shake to dissolve, then add $5\,\mathrm{mL}$ of $2\,\mathrm{mol/L}$ acetic acid-sodium acetate buffer solution, 5 drops of PAN indicator, and titrate slowly with $0.02035\,\mathrm{mol/L\,EDTA}$ until the endpoint is reached when the purple-red color changes to gray-blue-green, consuming $13.94\,\mathrm{mL}$.

(c) Transfer $10.00\,\mathrm{mL}$ of the $100\,\mathrm{mL}$ solution prepared in (b) to a $100\,\mathrm{mL}$ volumetric flask and dilute to the mark with distilled water. Transfer $25.00\,\mathrm{mL}$ of the solution to a small beaker, add $7\,\mathrm{mL}$ of $1:1$ ammonia water while stirring, heat for a period of time, and filter under normal pressure after the precipitate is obviously agglomerated. Wash the precipitate three times with deionized water. Discard the filtrate. Dissolve the precipitate with $15\,\mathrm{mL}$ of $0.4\mathrm{mol/L}$ sulfuric acid, collect the solution in a conical flask, add $10\,\mathrm{mL}$ of deionized water, 5 drops of sulfosalicylic acid indicator, and titrate with $0.008140\,\mathrm{mol/L\,EDTA}$ until the purple-red color fades significantly, which is the endpoint, consuming $14.53\,\mathrm{mL}$.

For this experiment, the following options are correct (for options involving calculations, consider errors within $0.1\ \%$ as correct)

A. Ascorbic acid thiourea can be used to mask Cu to measure Fe

B. Cyanide masking of Cu can be used to measure Fe.

C. Sulfur ion masking of Cu can be used to measure Fe.

D. Hydrochloric acid + hydrogen peroxide can be used as a complete replacement for nitric acid + hydrochloric acid.

E. (a) In experiments, high-temperature drying can be used to speed up the precipitation process.

F. (a) If there is no holding period at a constant temperature during the experiment, the calculated mass fraction of Se will be higher.

G. (c) The purpose of adding ammonia in the experiment is to adjust the pH to slightly acidic.

H. (c) The titration temperature in the experiment was room temperature.

I. Representing the alloy with $\mathrm{Fe}_x\mathrm{Cu}_y\mathrm{Se}$, then $x + y = 0.8969$

J. The ratio of ferrous iron to ferric iron is approximately $2.516 : 1$.

K. The mass fraction of Se is 1.954 times that of Fe.

L. All of the above options are incorrect.

Answer

Correct Answer: J

Detailed Explanation

Ascorbic acid thiourea, cyanide ions, and sulfide ions masking Cu will all react with Fe^{3+}, so they cannot be used, therefore options A, B, and C are all incorrect.

CHECKPOINT 1 PTS

Because ascorbic acid thiourea, cyanide ions, and sulfide ions all react with iron, options A, B, and C are all incorrect

Hydrogen peroxide decomposes upon heating, and Fe^{3+} catalyzes the decomposition of hydrogen peroxide, consuming too much hydrogen peroxide; and a small amount of nitric acid must be added to completely dissolve the sample, so it cannot be replaced, option D is incorrect.

CHECKPOINT 1 PTS

Because hydrogen peroxide is catalyzed by iron and decomposes upon heating, and nitric acid needs to be added to dissolve the sample, option D is incorrect

(a) In the experiment, high-temperature drying of the precipitate will cause Se to sublime, resulting in less precipitate obtained, option E is incorrect.

CHECKPOINT 1 PTS

High-temperature drying of the precipitate causes selenium to sublime, losing precipitate, option E is incorrect

(a) In the experiment, if it is not kept warm for a period of time, the precipitate will not be sufficiently aged, resulting in less precipitate obtained, and the calculated mass fraction of Se will be lower, option F is incorrect.

CHECKPOINT 0.5 PTS

Incomplete aging of the precipitate will result in loss of precipitate, option F is incorrect

CHECKPOINT 0.5 PTS

The calculated mass fraction of Se will be lower, option F is incorrect

(c) In the experiment, ammonia is added to adjust the pH to slightly alkaline to completely precipitate Fe. A slightly acidic environment will result in less precipitate obtained, option G is incorrect.

CHECKPOINT 0.5 PTS

Iron needs to precipitate in a slightly alkaline environment, option G is incorrect

(c) In the experiment, the titration temperature is $60-70$ degrees Celsius, because the reaction rate of EDTA and Fe is very slow at room temperature, causing the indicator to become sluggish. Therefore, the temperature is raised to increase the reaction rate of EDTA and Fe and avoid the indicator becoming sluggish, option H is incorrect.

CHECKPOINT 1 PTS

The reaction rate of EDTA and Fe is very slow at room temperature, causing the indicator to become sluggish, option H is incorrect

(a) Hydroxylamine hydrochloride precipitates Se, so the content of Se is determined by gravimetric analysis.

CHECKPOINT 1 PTS

(a) Hydroxylamine hydrochloride precipitates Se

The weight of selenium in the alloy sample is $m_{\mathrm{Se}} = 0.2565\ \mathrm{g}$

The mass fraction of selenium in the alloy sample is $\omega_{\mathrm{Se}} = \dfrac{0.2565\ \mathrm{g}}{0.4639\ \mathrm{g}} = 55.2921\ \%$

CHECKPOINT 0.5 PTS

$\omega_{Se} = 55.29\ \%$

(b) Ammonium fluoride masks Fe^{3+}, so the content of Cu is obtained by EDTA titration.

CHECKPOINT 1 PTS

(b) Ammonium fluoride masks Fe^{3+}, so the content of Cu is obtained by $EDTA$ titration.

The amount of substance of $EDTA$ consumed is $n_{EDTA} = 0.02035\ \mathrm{mol/L} \times 13.94\ \mathrm{mL} = 0.000283679\ \mathrm{mol}$

$EDTA$ and Cu^{2+} are complexed in a $1:1$ ratio, then in the titration solution $n_{Cu} = n_{EDTA} = 0.000283679\ \mathrm{mol}$

The amount of substance of copper in the alloy sample is $n_{Cu,total} = 4n_{Cu} = 0.00113472\ \mathrm{mol}$

The mass of copper in the alloy sample is $m_{Cu} = 0.00113472\ \mathrm{mol} \times 63.55\ \mathrm{g/mol} = 0.0721112\ \mathrm{g}$

The mass fraction of copper in the alloy sample is $\omega_{Cu} = \dfrac{0.0721112\ \mathrm{g}}{0.8527\mathrm{g}} = 8.45681\ \%$

CHECKPOINT 0.5 PTS

$\omega_{Cu} = 8.457\ \%$

(c) Fe^{3+} is precipitated by ammonia and then dissolved in sulfuric acid, so the content of Fe is obtained by $EDTA$ titration.

CHECKPOINT 1 PTS

(c) Fe^{3+} is precipitated by ammonia and then dissolved in sulfuric acid, so the content of Fe is obtained by $EDTA$ titration.

The amount of substance of $EDTA$ consumed is $n_{EDTA} = 0.008140\ \mathrm{mol/L} \times 14.53\ \mathrm{mL} = 0.000118274\ \mathrm{mol}$

EDTA and Cu^{2+} are complexed in a $1:1$ ratio, then in the titration solution $n_{\mathrm{Fe}} = n_{\mathrm{EDTA}} = 0.000118274\ \mathrm{mol}$

The amount of substance of iron in the alloy sample is $n_{\mathrm{Fe,total}} = 4 \times 10 \times n_{\mathrm{Fe}} = 0.00473097\ \mathrm{mol}$

The mass of iron in the alloy sample is $m_{\mathrm{Fe}} = 0.00473097\ \mathrm{mol} \times 55.85\ \mathrm{g/mol} = 0.264225\ \mathrm{g}$

The mass fraction of iron in the alloy sample is $\omega_{\mathrm{Fe}} = \dfrac{0.264225\ \mathrm{g}}{0.8527\ \mathrm{g}} = 30.9868\ \%$

CHECKPOINT 0.5 PTS

$\omega_{\mathrm{Fe}} = 30.99\ \%$

$\dfrac{\omega_{\mathrm{Se}}}{\omega_{\mathrm{Fe}}} = \dfrac{55.2921\ \%}{30.9868\ \%} = 1.78437$, so the mass fraction of Se is 1.784 times that of Fe, option K is incorrect.

CHECKPOINT 2 PTS

The mass fraction of Se is 1.784 times that of Fe, option K is incorrect

Let the alloy be represented by $\mathrm{Fe}_x\mathrm{Cu}_y\mathrm{Se}$, then $x = \dfrac{30.9868\ \%}{55.2921\ \%} \times \dfrac{78.96\ \mathrm{g/mol}}{55.85\ \mathrm{g/mol}} = 0.792315$

$y = \dfrac{8.45681\ \%}{55.2921\ \%} \times \dfrac{78.96\ \mathrm{g/mol}}{63.55\ \mathrm{g/mol}} = 0.190036$

Thus $x + y = 0.982351$, option I is incorrect.

CHECKPOINT 2 PTS

$x + y = 0.982351$, option I is incorrect

Let the ratio of ferrous iron to ferric iron be $z:1$, then the valence balance is $0.792315 \times \frac{2z+3}{z+1} + 0.190036 \times 1 + (-2) = 0$, solving for $z = 2.51618$

That is, the ratio of ferrous iron to ferric iron is approximately $2.516:1$, option J is correct.

CHECKPOINT 2 PTS

The ratio of ferrous iron to ferric iron is approximately $2.516:1$, option J is correct

Therefore, choose J.

CHECKPOINT 1 PTS

Option J is correct